Greg Chism

From social insects to machine learning: data science grounded in messy, real-world systems.

Assistant Professor of Practice · gchism@arizona.edu
College of Information Science · University of Arizona

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Beyond the desk

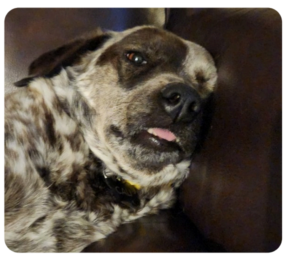
Curiosity about how small agents build big structures is the through-line. It started with animals and never really stopped.

National parks. The long walk-in, the early start.

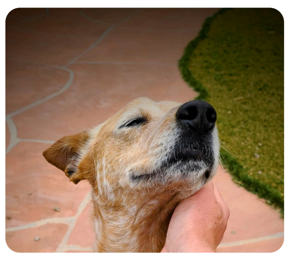
Fly fishing. Mostly an excuse to stand in cold water and think.

The crew. Three dogs with three incompatible opinions about the office chair.

Finn

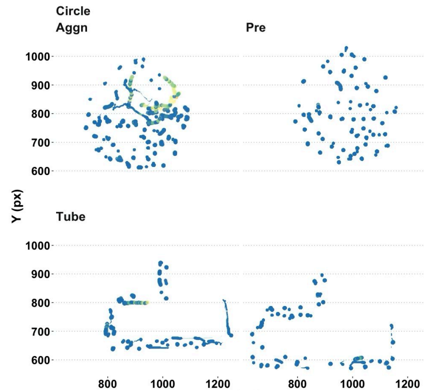


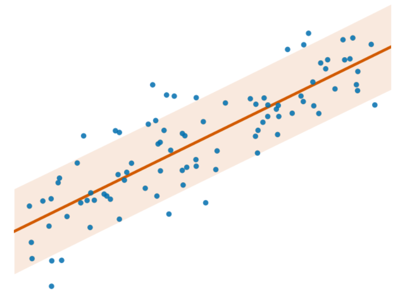
Cali



Miles

How I got here





Social spiders

UCSB · Undergraduate

Social insects

Arizona · Ph.D.

Nest analytics

Where the data turn began

Data science

Arizona · College of Information Science

The turn: measuring *how ant colonies use the space they build* meant tracking hundreds of individuals for days at a time. The bottleneck stopped being the microscope and became the analysis, so I went to fix the analysis and never came back.

Two research programs

01 · Healthcare analytics: measurement you can trust

02 · AI-assisted research workflows: pipelines that scale

Healthcare: the question

Does chronic pain distort depression screening?

The PHQ-8 counts somatic symptoms: **sleep, fatigue, appetite, psychomotor slowing**. Those are also everyday features of chronic pain, which raises the worry of *criterion contamination*. The tool might be counting pain and recording it as depression.

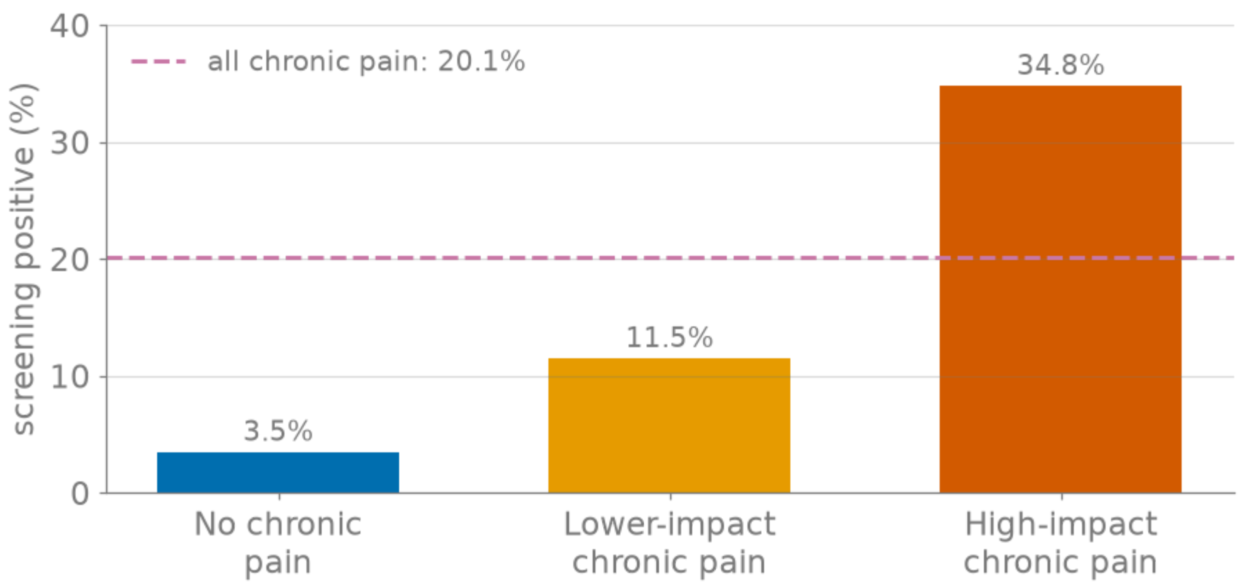


Figure 1: Positive PHQ-8 screens (≥10) by pain impact, 2019 NHIS (n = 30,983). Dashed line: all chronic pain combined (20.1%), which is the union of the two CP bars rather than a fourth group.

The finding: it doesn't. The gap is real rather than an artifact of the instrument.

Healthcare: why it's trustworthy

Check	No CP	CP
Cronbach's α	0.82	0.85
McDonald's ω	0.86	0.89
CFI	0.994	0.993
TLI	0.991	0.991

Invariance holds at the configural, metric, **and scalar** levels. **No differential item functioning.**

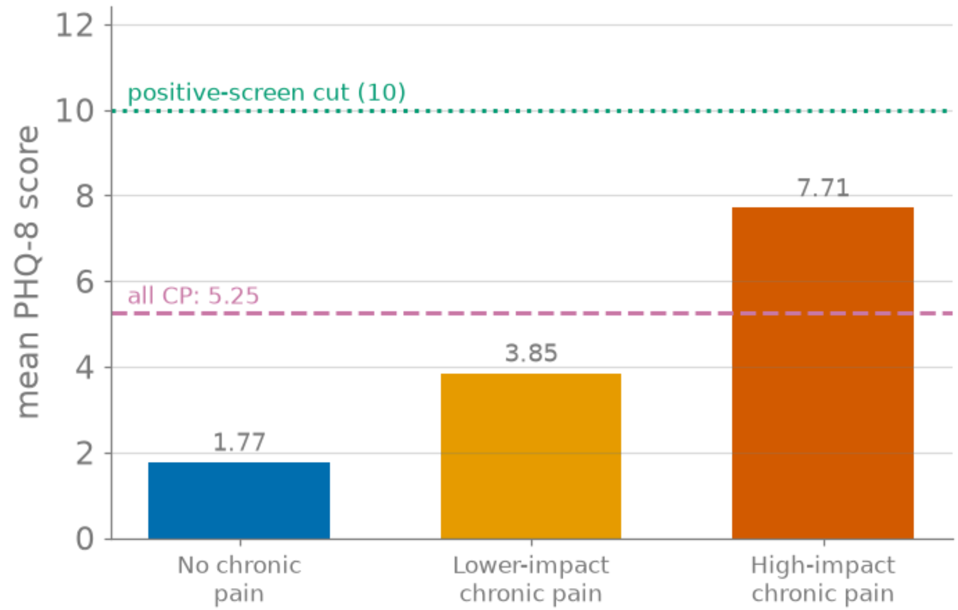


Figure 2: Mean PHQ-8 by pain impact. Even the high-impact mean (7.71) sits below the ≥10 screening cut point.

Bottom line: the PHQ-8 measures the *same thing* in both groups, so cross-group comparison is valid, and the 5-to-10× disparity is a real clinical difference rather than a measurement artifact.

AI workflows: the system

3,681 state bills (OpenStates + LegiScan, 2017–) → **2,364 AI-relevant** (64%), 51 jurisdictions, **14-category** governance taxonomy, mean 2.86 categories per bill.

Six-model ensemble, zero-shot. Three Claude models (Sonnet, Opus, Haiku) and three open-source (Qwen-2.5-72B, Llama-3.1-70B/8B). F1-weighted majority vote, with confidence tiers from raw six-model agreement.

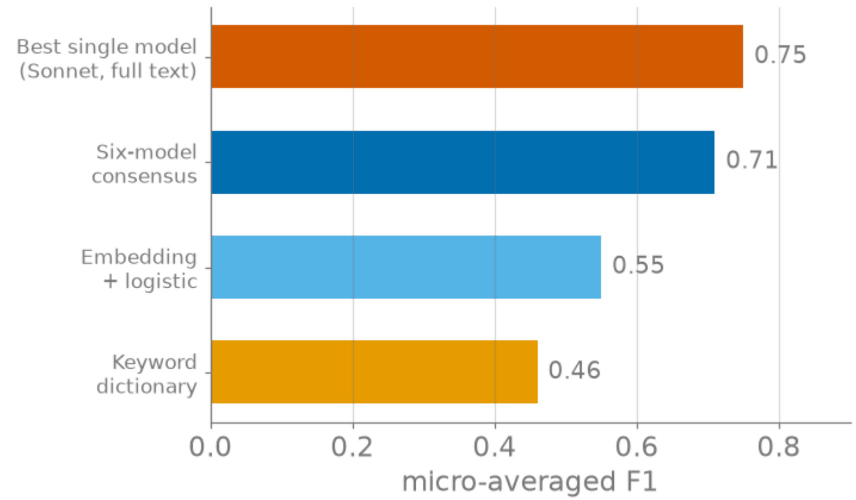


Figure 3: F1 on the n = 20 human gold set. The ensemble does *not* beat Sonnet alone. What it buys is calibrated confidence.

Honest caveat: the gold set is **n = 20, single-coder**. That fixes the *ordering* of methods, but not tight per-category intervals.

AI workflows: what the data shows

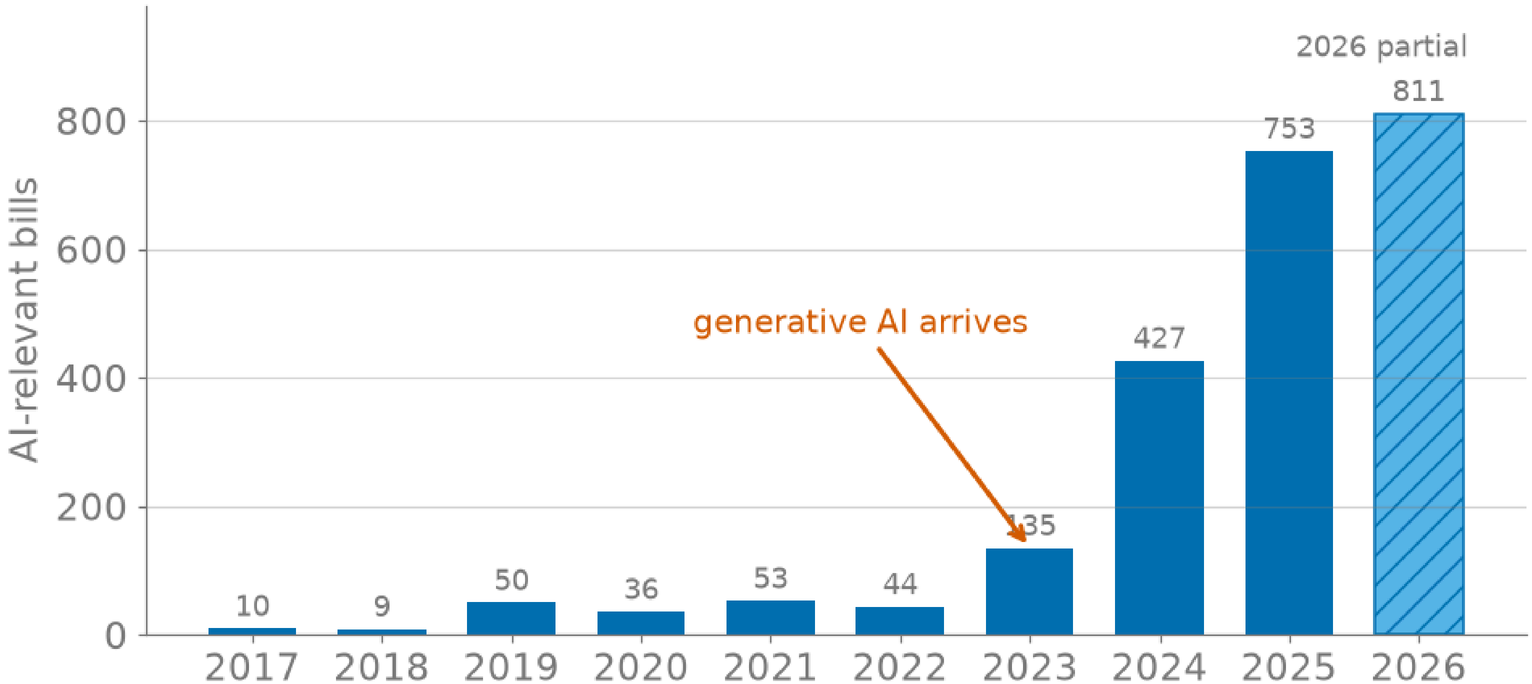


Figure 4: AI-relevant bills by year of first action. 2026 (hatched) is a partial year, with the corpus frozen 2026-07-10, and it has already passed all of 2025.

Enactment: 17.7% of 1,935 decided bills. Narrow bills pass: Education 22.9%, Criminal Use 22.8%. Accountability stalls: Private Right of Action 12.8%, Industry Use 12.8%, Taxes 5.7%.

Cox model (2,250 bills, 342 events): sponsor count **HR 1.23**; Disclosure **HR 0.80** and Private Right of Action **HR 0.79** *slow* enactment. The mechanisms with teeth are the ones that stall.

INFO 521

Probabilistic and from-scratch · healthcare-grounded · reproducible by design

Learning outcomes

By the end of this course, you will be able to:

1. **Identify and evaluate** machine-learning approaches appropriate for regression, classification, clustering, and dimensionality-reduction problems.
2. **Implement** machine-learning algorithms in Python using industry-standard tools and workflows.
3. **Compare and interpret** model performance using appropriate evaluation metrics, validation techniques, and statistical reasoning.
4. **Apply** machine-learning methods to real-world datasets to develop predictive and descriptive models.
5. **Communicate** results, limitations, and recommendations through written, visual, and technical presentations.

The seven-module arc

loss → likelihood → posterior → approximation → discovery

	Module	The idea
M1	Linear Least Squares	the model, the normal equations
M2	Generalization	CV, bias–variance, ridge
M3	Likelihood & MLE	least squares, re-earned
M4	Bayesian Inference	priors, posteriors, conjugacy
M5	Approximate Inference	logistic + Newton, Laplace, MCMC
M6	SVM & Clustering	max-margin, k-means, GMM + EM
M7	PCA	eigen-decomposition of the covariance

One object runs through all of it: $X^\top X$. The least-squares bowl (M1), the likelihood's curvature and Fisher information (M3), the Laplace posterior covariance (M5), the covariance you eigen-decompose (M7).

How the course works

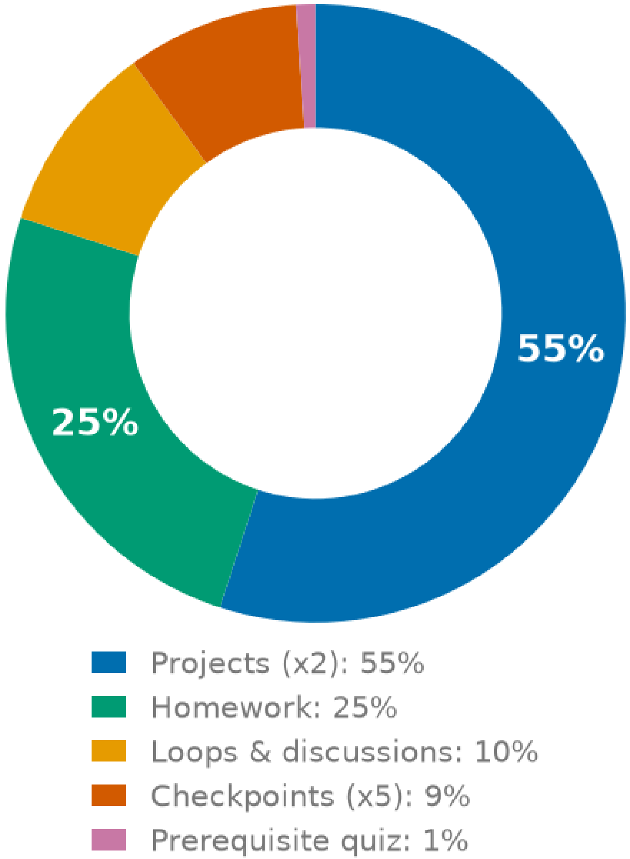


Figure 5: Grade weighting. Projects and homework together make up 80%, the work that looks most like real practice.

- **Specs grading.** Work is **Pass** or **Not-Yet**, never partial credit. Autograded correctness resubmits **without limit**; human-reviewed components draw on a small pool of revision tokens.
- **Two projects, one seam.** *One Problem, Three Lenses* → *From Inference to Discovery*. Part 1 closes on the **conjugate Gaussian posterior**; Part 2 opens by **binarizing the outcome to break conjugacy**, which is exactly what forces approximate inference.
- **Interactive weekly activities.** Each module ships a browser tool you use in lecture and again in the peer loop.
- **Peer-engagement loops, four stages:** tool + mastery check → reflection → review **every** teammate → respond and revise. Credit needs all four.

Where this lands

The clinical data you will model in INFO 521 is **the same kind of data I study**: national health survey data, self-reported instruments, real measurement error, and consequences attached to the answer.

So the questions this course trains you to ask are not academic:

- Is the gap I measured **real**, or an artifact of the instrument?
- Does my model do better than the **simplest thing that could work**?
- What does my method get wrong, and **who does that cost**?

Every method in this course is a way of being **honest about uncertainty**.

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